

PP1 PANEL PREPARATION GUIDE

Project R26-IT-117 — AI-Based Architectural Planning Model

Component 01 — Architectural Planning API

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1. PROJECT OVERVIEW

What it does: Takes a Sri Lankan cadastral (survey) plan as input and automatically generates 3 floor plan alternatives as SVG/PDF output, following NBC Sri Lanka building regulations.

Input: Cadastral plan (PDF/JPG/PNG), user requirements (rooms, floors, style, budget)

Output: 3 floor plan alternatives (SVG + PDF), Building Schema JSON for Component 02

2. THE 4-STAGE PIPELINE

Stage 1 — Cadastral Plan Extraction | POST /api/v1/extract | Synchronous

- Uploads a cadastral plan image or PDF
- EasyOCR reads all text from the plan
- OpenCV detects the plot boundary polygon
- spaCy fine-tuned NER extracts: plan number, district, area, scale, surveyor, road access, coordinates (F=0.857, trained on 69 real SL plans)
- pyproj converts SLD99 survey coordinates to WGS84 GPS
- Output: validated Site Schema JSON

Stage 2 — Buildable Zone Calculation | POST /api/v1/buildable-zone | Synchronous

- Applies NBC Sri Lanka setbacks (front/rear/side margins)
- Applies BCR (Building Coverage Ratio) based on district and coastal flag
- Shapely computes the legal buildable polygon
- Output: buildable polygon, max footprint sqm, recommended floors, orientation

Stage 3 — AI Floor Plan Generation | POST /api/v1/generate-floor-plan | Async (Celery)

- RAG retrieves 5 similar Sri Lankan house designs from Supabase pgvector (131 entries)
- PromptBuilder constructs structured prompt with site constraints and RAG context
- Gemini 2.5 Flash generates 3 alternatives at temperatures 0.4 / 0.7 / 1.0
- Strip-packing layout solver eliminates room overlaps — all plans return is_valid=true
- Validator checks geometric violations, Scorer rates each plan on 4 dimensions
- Returns task_id immediately — poll GET /api/v1/floor-plan-status/{task_id}

Stage 4 — Render SVG/PDF | POST /api/v1/render | Synchronous

- SVGRenderer draws floor plan with thick walls, door arcs, furniture symbols, staircase pattern, dimension lines, north arrow, title block
- PDFRenderer generates 6-page professional report (ReportLab)
- SchemaSerialiser validates Building Schema and POSTs to Component 02
- Output: SVG file + PDF file + Building Schema JSON

3. MODELS & TECHNOLOGIES

Model / Library	Stage	Purpose
EasyOCR (CRAFT + CRNN)	Stage 1	Reads text from cadastral plan images
PyTorch CNN (ResNet-based)	Stage 1	Classifies cadastral vs non-cadastral plans
OpenCV (Canny + contours)	Stage 1	Detects plot boundary polygon
spaCy fine-tuned NER	Stage 1	Custom NER trained on 69 real SL plans, F=0.857
pyproj (EPSG:5234 to 4326)	Stage 1	Converts SLD99 to WGS84 GPS coordinates
Shapely	Stage 2	Geometric polygon operations
NBC Rule Engine (custom)	Stage 2	Sri Lanka National Building Code setbacks
all-MiniLM-L6-v2 (SentenceTransformer)	Stage 3	384-dim embeddings for RAG vector search
Supabase pgvector	Stage 3	Cloud vector database — 131 floor plan entries
Gemini 2.5 Flash (Google)	Stage 3	LLM — generates 3 floor plan alternatives
layout_solver.py (strip packing)	Stage 3	Post-processing solver eliminates room overlaps
svgwrite	Stage 4	Draws SVG with walls, doors, furniture, staircase
ReportLab	Stage 4	Generates 6-page professional PDF report
jsonschema	Stage 4	Validates Building Schema JSON
FastAPI	API	REST API framework
Celery + Redis	Stage 3	Async task queue for Gemini generation
Docker	Infra	Containerisation for deployment

4. DATASET & DATA COLLECTION

Component	Dataset	Status
EasyOCR	Pre-trained multilingual model (ICDAR, SynthText)	Ready
CNN Classifier	Collecting 100+ SL cadastral plans from licensed surveyors	In progress
NER Parser	Fine-tuned on 69 real SL cadastral plans + 41 synthetic	F=0.857 — completed
RAG Knowledge Base	121 CVC FPP floor plan images via Gemini Vision + 10 original	131 entries in Supabase
Gemini 2.5 Flash	Google pre-trained LLM	Ready — API integrated

CNN Classifier — 5 Classes

- **surveyor_plan** — Standard licensed surveyor cadastral plan
- **municipal_plan** — Municipal council subdivision plan
- **uda_plan** — Urban Development Authority plan
- **low_quality** — Scan too poor for reliable extraction
- **non_cadastral** — Not a cadastral plan — reject

5. ACCURACY & PERFORMANCE METRICS

Component	Metric	Value
EasyOCR	Text recognition on printed cadastral plans	85–95%

Component	Metric	Value
CNN Classifier	5-class classification accuracy	Pending — expected 90%+
NER Parser (fine-tuned)	F-score on held-out eval set	0.857 (P=0.941, R=0.787)
Coordinate Conversion	SLD99 to WGS84	Mathematically exact (EPSG:5234)
RAG Retriever	Passages retrieved per query	5 passages, cosine similarity
Floor Plan Validity	is_valid=true rate after layout solver	100% (3/3 alternatives)
Floor Plan Scorer	Quality dimensions	4: space utilisation, natural light, adjacency, ver
Celery Task	End-to-end generation time	60–170 seconds
Test Suite	Unit + integration tests	31/31 passing

6. PANEL QUESTIONS & ANSWERS

Q: What AI/ML processes are in your system?

- 1. Computer Vision — OpenCV edge detection for boundary polygon
- 2. Deep Learning OCR — EasyOCR CRAFT+CRNN neural network
- 3. CNN Classification — ResNet-based cadastral plan classifier
- 4. NLP/NER — Fine-tuned Named Entity Recognition (F=0.857)
- 5. RAG — vector similarity search combined with LLM generation
- 6. LLM Prompting — Gemini 2.5 Flash with structured prompt engineering
- 7. Geometric Validation + Solver — Shapely polygon intersection + strip-packing layout solver

Q: You use pre-trained models — is that allowed?

Yes. We use pre-trained models as a base and extend them for the Sri Lankan domain. EasyOCR is extended with Sri Lankan cadastral text patterns. spaCy base model is fine-tuned on 69 real Sri Lankan cadastral plans achieving F=0.857. CNN classifier is being fine-tuned on 100+ real plans currently being collected from licensed surveyors. Gemini is prompted with Sri Lankan NBC regulations and RAG context from 131 local house design entries.

Q: What dataset did you use?

- 1. EasyOCR — pre-trained, no dataset needed
- 2. CNN — collecting 100+ SL cadastral plans from licensed surveyors (father and colleagues)
- 3. NER — fine-tuned on 69 real cadastral plans from Ampara district surveyors + 41 synthetic. F=0.857.
- 4. RAG — 131 entries: 10 original SL coastal descriptions + 121 CVC FPP floor plans via Gemini Vision

Q: What is your accuracy?

EasyOCR: 85–95% on printed cadastral text. NER fine-tuned model: F=0.857 (P=0.941, R=0.787) on real cadastral plans. Coordinate conversion: mathematically exact. CNN: fine-tuning in progress, expected 90%+. Floor plan validity: 100% after layout solver. 31/31 unit tests passing.

Q: Why did you use RAG?

RAG gives Gemini real Sri Lankan house design context. Without RAG, Gemini generates generic Western-style floor plans. With RAG, it generates designs suited for Sri Lankan climate — cross-ventilation, shade, coastal pile foundations, verandahs. We built our RAG store from 121 real floor plan images (CVC FPP dataset) analysed by Gemini Vision, giving 131 total entries in Supabase pgvector. RAG retrieves the 5 most similar designs using cosine similarity on 384-dimensional sentence embeddings.

Q: Why 3 temperature values for Gemini?

Temperature controls creativity in LLM output:

- 0.4 (Conservative) — low randomness, strictly follows NBC constraints
- 0.7 (Balanced) — moderate creativity, balances rules and innovation
- 1.0 (Creative) — high randomness, more innovative designs
- This gives the user 3 real alternatives to choose from.

Q: Why Celery and Redis?

Gemini takes 60–170 seconds to generate 3 floor plans. A synchronous HTTP request would time out. Celery runs the generation as a background task, the user gets a task_id immediately and polls /floor-plan-status/{task_id} for the result. Redis is the message broker and result backend.

Q: Why Supabase instead of a local vector database?

Supabase pgvector is cloud-hosted PostgreSQL with vector extension — no local ChromaDB to manage or deploy. Works from anywhere, persistent storage, and integrates with standard SQL for structured queries alongside vector search.

Q: What is SLD99 and why do you convert it?

SLD99 (EPSG:5234) is Sri Lanka's official local coordinate system used by all licensed surveyors. GPS uses WGS84 (EPSG:4326). We convert so the plot location can be displayed on Google Maps and used by other system components. pyproj handles the mathematical transformation. Example: SLD99 E=319950, N=231025 converts to GPS lat=7.280069, lon=81.859983 (Ampara district).

Q: What NBC regulations do you follow?

NBC = National Building Code of Sri Lanka.

- Front setback: 3m (national road), 2m (provincial), 1.5m (local), 1m (lane)
- Rear setback: 1.5m minimum
- Side setback: 1m minimum
- BCR: Colombo 0.60, Coastal districts 0.40 (stricter), Other districts 0.50
- Coastal sites require pile foundations

Q: How does your component connect to other components?

Input: Receives cadastral plan upload from frontend (Component 04). Output: Sends Building Schema JSON to Component 02 (cost estimation) via HTTP POST to /estimate endpoint. If Component 02 is unavailable, Stage 4 logs a warning and continues — the render is never blocked by C02 availability.

Q: What are the limitations?

- 1. CNN fine-tuning in progress — collecting real Sri Lankan cadastral plans
- 2. Room overlap issue resolved via strip-packing layout solver — all plans now return is_valid=true
- 3. Currently tested primarily on Ampara district data — expanding to all 25 districts
- 4. Generation takes 60–170 seconds due to 3 Gemini API calls

7. NBC SRI LANKA QUICK REFERENCE

Road Type	Front Setback	Rear Setback	Side Setback
National Road / Main Road	3.0m	1.5m	1.0m
Provincial Road	2.0m	1.5m	1.0m
Local Road	1.5m	1.5m	1.0m
Lane / Private Road	1.0m	1.5m	1.0m

Building Coverage Ratio (BCR)

District Type	BCR	Meaning
Colombo (urban)	0.60	60% of land can be built
Other Districts (standard)	0.50	50% of land can be built
Coastal Districts (Ampara, Galle, etc.)	0.40	40% of land — stricter

Foundation Type Rules

- Coastal site — Pile foundation

- Large plot (area > 500 sqm) — Raft foundation
- Standard plot — Strip foundation

8. API ENDPOINTS

Method	Endpoint	Description
POST	/api/v1/extract	Stage 1 — Upload cadastral plan, get Site Schema
POST	/api/v1/buildable-zone	Stage 2 — Compute buildable polygon
POST	/api/v1/generate-floor-plan	Stage 3 — Dispatch async floor plan generation
GET	/api/v1/floor-plan-status/{task_id}	Stage 3 — Poll for generation result
POST	/api/v1/render	Stage 4 — Render SVG + PDF + Building Schema
GET	/api/v1/download/{filename}	Download rendered SVG or PDF file
GET	/api/v1/health	Health check

Infrastructure

- API Server: FastAPI + Uvicorn (port 8001)
- Task Queue: Celery 5.4 + Redis
- Vector DB: Supabase pgvector (131 entries)
- LLM: Gemini 2.5 Flash (Google AI Studio)
- Container: Docker

9. DEMO FLOW FOR PANEL

- Step 1** — GET /api/v1/health — verify status:ok, component:C01
- Step 2** — POST /api/v1/extract with cadastral plan image — get Site Schema JSON (plan_id, district, area_sqm, GPS coords)
- Step 3** — POST /api/v1/buildable-zone — get buildable polygon, max footprint, setback margins
- Step 4** — POST /api/v1/generate-floor-plan — get task_id immediately (async dispatch)
- Step 5** — GET /api/v1/floor-plan-status/{task_id} — poll until DONE — 3 alternatives: conservative, balanced, creative — all is_valid=true
- Step 6** — POST /api/v1/render with chosen plan — get SVG + PDF + Building Schema JSON
- Step 7** — Open SVG file — show floor plan with thick walls, door arcs, furniture symbols, room labels, north arrow, title block

KEY NUMBERS TO REMEMBER

- ✓ 131 RAG entries in Supabase (10 original + 121 from CVC FPP dataset)
- ✓ 69 real cadastral plans for NER training + 41 synthetic = 110 total
- ✓ NER F=0.857, Precision=0.941, Recall=0.787
- ✓ 3 floor plan alternatives per generation (conservative / balanced / creative)
- ✓ 100% validity rate after strip-packing layout solver
- ✓ 31/31 unit tests passing
- ✓ 60–170 seconds generation time
- ✓ 7 AI/ML processes in the system